Comprehensive Exam – PDE2, Top 10– Spring 2026
w/Faculty

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Here is a ***Top 10 PDE Comprehensive Exam Practice Set***, designed around the kinds of problems that repeatedly appear on graduate qualifying/comprehensive exams.

These focus on the "core moves" of graduate PDE:

- weak formulations,
- energy estimates,
- Sobolev embeddings,
- compactness,
- maximum principles,
- Fourier methods,
- variational methods.

These are the backbone of graduate Partial Differential Equations.

1. Energy estimate for the heat equation

Let u solve

$$u_t - \Delta u = 0 \quad \text{in } U \times (0,T),$$

with $u = 0$ on $\partial U \times (0,T)$.

Show that

$$\frac{d}{dt}\int_U u^2\,dx \le 0.$$

Then prove uniqueness.

Skills tested:

- multiply by u,
- integration by parts,
- energy decay.

We use the standard ***energy method*** for the heat equation.

Let u solve

$$u_t - \Delta u = 0 \qquad \text{in } U \times (0,T),$$

with boundary condition

$$u = 0 \qquad \text{on } \partial U \times (0,T).$$

We will show:

$$\frac{d}{dt}\int_U u^2\,dx \le 0,$$

and then use this to prove uniqueness.

This is the basic energy estimate for the heat equation in Partial Differential Equations.

Step 1: Multiply the PDE by u

Starting with

$$u_t - \Delta u = 0,$$

multiply by u:

$$uu_t - u\Delta u = 0.$$

Integrate over U:

$$\int_U uu_t\,dx - \int_U u\Delta u\,dx = 0.$$

Step 2: Integrate by parts

Using integration by parts,

$$\int_U u\Delta u\,dx = -\int_U |\nabla u|^2\,dx + \int_{\partial U} u\frac{\partial u}{\partial \nu}\,dS.$$

Since $u = 0$ on ∂U, the boundary term vanishes:

$$\int_{\partial U} u\frac{\partial u}{\partial \nu}\,dS = 0.$$

Thus,

$$\int_U u\Delta u\,dx = -\int_U |\nabla u|^2\,dx.$$

Substitute back:

$$\int_U uu_t\,dx + \int_U |\nabla u|^2\,dx = 0.$$

Step 3: Recognize the energy derivative

Since

$$\frac{d}{dt}\int_U u^2\,dx = 2\int_U uu_t\,dx,$$

we obtain

$$\frac{1}{2}\frac{d}{dt}\int_U u^2\,dx + \int_U |\nabla u|^2\,dx = 0.$$

Therefore,

$$\frac{d}{dt}\int_U u^2\,dx = -2\int_U |\nabla u|^2\,dx.$$

Hence,

$$\boxed{\frac{d}{dt}\int_U u^2\,dx \le 0.}$$

So the L^2-energy is nonincreasing.

Step 4: Prove uniqueness

Suppose u_1, u_2 are two solutions with the same initial data.

Define

$$w = u_1 - u_2.$$

Then w satisfies

$$w_t - \Delta w = 0$$

in $U \times (0, T)$, with

$$w = 0$$

on $\partial U \times (0, T)$, and initial condition

$$w(x, 0) = 0.$$

Apply the energy identity to w:

$$\frac{d}{dt}\int_U w^2\,dx \le 0.$$

But initially,

$$\int_U w(x,0)^2\,dx = 0.$$

Since the energy is nonincreasing and nonnegative,

$$\int_U w(x,t)^2\,dx = 0 \qquad \forall t.$$

Thus,

$$w(x,t) = 0 \qquad \forall (x,t).$$

Therefore,

$$u_1 = u_2.$$

Hence the solution is unique.

Final result

We proved:

$$\boxed{\frac{d}{dt}\int_U u^2\,dx = -2\int_U |\nabla u|^2\,dx \le 0}$$

and this implies:

$$\boxed{\text{The solution of the heat equation is unique.}}$$

Interpretation

The quantity

$$\int_U u^2\,dx$$

is the heat energy, and diffusion causes it to decrease over time.

That decay prevents two different solutions from starting from the same initial data.

If you'd like, next I can show how this energy estimate yields the ***stability estimate***

$$|u(\cdot,t)| * L^2(U) \le |g| * L^2(U),$$

which is another common comprehensive exam result.

2. Energy conservation for the wave equation

Let u solve

$$u_{tt} - \Delta u = 0$$

with homogeneous Dirichlet boundary data.

Show that

$$E(t) = \frac{1}{2}\int_U (u_t^2 + |\nabla u|^2)\, dx$$

is constant.

Skills tested:

- multiply by u_t,
- conserved energy,
- uniqueness.

We prove that the energy

$$E(t) = \frac{1}{2}\int_U \left(u_t^2 + |\nabla u|^2\right)\, dx$$

is constant for solutions of

$$u_{tt} - \Delta u = 0 \qquad \text{in } U \times (0,T),$$

with homogeneous Dirichlet boundary condition

$$u = 0 \qquad \text{on } \partial U \times (0,T).$$

This is the standard ***energy conservation law*** for the wave equation in Partial Differential Equations.

Step 1: Differentiate the energy

Differentiate $E(t)$:

$$E'(t) = \frac{1}{2}\frac{d}{dt}\int_U \left(u_t^2 + |\nabla u|^2\right)\, dx.$$

So

$$E'(t) = \int_U u_t u_{tt}\, dx + \int_U \nabla u \cdot \nabla u_t\, dx.$$

Step 2: Integrate by parts

Integrate the second term by parts:

$$\int_U \nabla u \cdot \nabla u_t\, dx = -\int_U (\Delta u) u_t\, dx + \int_{\partial U} \frac{\partial u}{\partial \nu} u_t\, dS.$$

Thus,

$$E'(t) = \int_U u_t u_{tt}\, dx - \int_U (\Delta u) u_t\, dx + \int_{\partial U} \frac{\partial u}{\partial \nu} u_t\, dS.$$

Factor u_t:

$$E'(t) = \int_U u_t(u_{tt} - \Delta u)\, dx + \int_{\partial U} \frac{\partial u}{\partial \nu} u_t\, dS.$$

Step 3: Use the PDE

Since

$$u_{tt} - \Delta u = 0,$$

the volume term vanishes:

$$\int_U u_t(u_{tt} - \Delta u)\, dx = 0.$$

Hence

$$E'(t) = \int_{\partial U} \frac{\partial u}{\partial \nu} u_t \, dS.$$

Step 4: Use the boundary condition

Since

$$u = 0 \qquad \text{on } \partial U \times (0, T),$$

we have

$$u_t = 0 \qquad \text{on } \partial U \times (0, T),$$

because the boundary value is identically zero in time.

Thus,

$$\int_{\partial U} \frac{\partial u}{\partial \nu} u_t \, dS = 0.$$

Therefore,

$$E'(t) = 0.$$

Step 5: Conclude conservation

Since

$$E'(t) = 0,$$

the energy is constant:

$$\boxed{E(t) = \frac{1}{2} \int_U \left(u_t^2 + |\nabla u|^2\right) dx \text{ is constant in time.}}$$

Final conclusion

For the wave equation with homogeneous Dirichlet boundary data,

$$\boxed{\frac{d}{dt}\left[\frac{1}{2}\int_U (u_t^2 + |\nabla u|^2)\, dx\right] = 0}$$

so the total energy is conserved.

Interpretation

The two terms represent:

$$\frac{1}{2}\int_U u_t^2 \, dx \quad \text{(kinetic energy)}$$

and

$$\frac{1}{2}\int_U |\nabla u|^2 \, dx \quad \text{(potential energy)}.$$

The wave equation transfers energy between these two forms, but the total remains constant.

—

If you'd like, next I can show how this same computation gives ***uniqueness of the wave equation***, exactly as the heat equation energy estimate gave uniqueness.

3. Weak formulation of Poisson's equation

Given

$$-\Delta u = f \quad \text{in } U, \qquad u = 0 \text{ on } \partial U,$$

derive the weak formulation:

$$\int_U \nabla u \cdot \nabla v \, dx = \int_U f v \, dx.$$

Skills tested:

- integration by parts,
- Sobolev spaces,
- weak derivatives.

To derive the ***weak formulation*** of

$$-\Delta u = f \quad \text{in } U, \qquad u = 0 \quad \text{on } \partial U,$$

we convert the PDE into an integral identity using ***test functions*** and ***integration by parts***. This is the standard weak formulation for Poisson's equation in Partial Differential Equations.

Step 1: Choose a test function

Let

$$v \in C_c^\infty(U),$$

that is, v is smooth and compactly supported in U.
Multiply the PDE by v:

$$-\Delta u \, v = f v.$$

Integrate over U:

$$\int_U (-\Delta u) v \, dx = \int_U f v \, dx.$$

Step 2: Integrate by parts

Use Green's identity:

$$\int_U (-\Delta u) v \, dx = \int_U \nabla u \cdot \nabla v \, dx - \int_{\partial U} v \frac{\partial u}{\partial \nu} \, dS.$$

Since $v \in C_c^\infty(U)$, we have

$$v = 0 \quad \text{on } \partial U,$$

so the boundary term vanishes:

$$\int_{\partial U} v \frac{\partial u}{\partial \nu} \, dS = 0.$$

Thus,

$$\int_U (-\Delta u) v \, dx = \int_U \nabla u \cdot \nabla v \, dx.$$

Substituting into the previous equation gives:

$$\int_U \nabla u \cdot \nabla v \, dx = \int_U f v \, dx.$$

Step 3: Extend to Sobolev test functions

The above identity holds for all

$$v \in C_c^\infty(U).$$

Since $C_c^\infty(U)$ is dense in $H_0^1(U)$, we extend the identity to all

$$v \in H_0^1(U).$$

Thus the weak formulation is:

$$\boxed{\int_U \nabla u \cdot \nabla v \, dx = \int_U f v \, dx \qquad \forall v \in H_0^1(U).}$$

Final answer

The weak formulation of

$$-\Delta u = f, \qquad u = 0 \text{ on } \partial U$$

is:

$$\boxed{\int_U \nabla u \cdot \nabla v \, dx = \int_U f v \, dx \qquad \forall v \in H_0^1(U).}$$

Why this is useful

The classical PDE requires second derivatives:

$$\Delta u.$$

But the weak formulation only requires first derivatives:

$$\nabla u.$$

So instead of needing $u \in C^2(U)$, we only need

$$u \in H_0^1(U),$$

which makes existence theory much easier.

If you'd like, next I can show how to use this weak formulation with ***Lax–Milgram*** to prove existence and uniqueness of u.

4. Existence by Lax–Milgram

Show there exists a unique weak solution to

$$-\Delta u + u = f, \qquad u|_{\partial U} = 0$$

for $f \in L^2(U)$.

Skills tested:

- bilinear form,
- boundedness,
- coercivity,
- Lax–Milgram.

We will prove existence and uniqueness of a weak solution using the ***Lax–Milgram theorem***, the standard method for elliptic problems in Partial Differential Equations.

Step 1: Write the weak formulation

We consider

$$-\Delta u + u = f \quad \text{in } U, \qquad u = 0 \quad \text{on } \partial U.$$

Multiply by a test function $v \in H_0^1(U)$:

$$(-\Delta u + u)v = fv.$$

Integrate over U:

$$\int_U (-\Delta u)v\,dx + \int_U uv\,dx = \int_U fv\,dx.$$

Integrate the Laplacian term by parts:

$$\int_U (-\Delta u)v\,dx = \int_U \nabla u \cdot \nabla v\,dx,$$

since $v = 0$ on ∂U.

Thus the weak formulation is:

$$\boxed{\int_U (\nabla u \cdot \nabla v + uv)\,dx = \int_U fv\,dx \qquad \forall v \in H_0^1(U).}$$

Step 2: Define the bilinear form

Define

$$a(u,v) = \int_U (\nabla u \cdot \nabla v + uv)\,dx.$$

Define the linear functional

$$L(v) = \int_U fv\,dx.$$

Then the weak problem is:

$$a(u,v) = L(v) \qquad \forall v \in H_0^1(U).$$

Now we verify the hypotheses of ***Lax–Milgram***.

Step 3: Show $a(\cdot,\cdot)$ is bounded

We estimate:

$$|a(u,v)| \leq \int_U |\nabla u||\nabla v|\,dx + \int_U |u||v|\,dx.$$

Using Cauchy–Schwarz:

$$|a(u,v)| \leq \|\nabla u\|_{L^2}\|\nabla v\|_{L^2} + \|u\|_{L^2}\|v\|_{L^2}.$$

Hence,

$$|a(u,v)| \leq \|u\|_{H_0^1(U)}\|v\|_{H_0^1(U)}.$$

So a is bounded.

Step 4: Show coercivity

Compute:

$$a(u,u) = \int_U (|\nabla u|^2 + u^2)\,dx.$$

Thus,

$$a(u,u) = \|u\|^2_{H_0^1(U)}.$$

Hence

$$a(u,u) \geq \|u\|^2_{H_0^1(U)}.$$

So a is coercive.

Step 5: Show L is bounded

We estimate:

$$|L(v)| = \left|\int_U fv\,dx\right| \leq \|f\|_{L^2}\|v\|_{L^2}.$$

Since

$$\|v\|_{L^2} \leq \|v\|_{H_0^1(U)},$$

we get

$$|L(v)| \leq \|f\|_{L^2}\|v\|_{H_0^1(U)}.$$

So L is a bounded linear functional.

Step 6: Apply Lax–Milgram

Since:

1. $a(u,v)$ is bounded,
2. $a(u,u)$ is coercive,
3. $L(v)$ is bounded,

the ***Lax–Milgram theorem*** gives a unique $u \in H_0^1(U)$ such that

$$a(u,v) = L(v) \qquad \forall v \in H_0^1(U).$$

That is,

$$\int_U (\nabla u \cdot \nabla v + uv)\,dx = \int_U fv\,dx \qquad \forall v \in H_0^1(U).$$

So there exists a unique weak solution.

Final conclusion

There exists a unique

$$u \in H_0^1(U)$$

such that

$$\boxed{\int_U (\nabla u \cdot \nabla v + uv)\, dx = \int_U fv\, dx \quad \forall v \in H_0^1(U),}$$

and this is the unique weak solution of

$$-\Delta u + u = f, \qquad u|_{\partial U} = 0.$$

Key structure

The proof is always:

1. derive weak form,
2. define bilinear form,
3. prove boundedness,
4. prove coercivity,
5. apply Lax–Milgram.

This is the standard existence/uniqueness framework for linear elliptic PDE.

If you'd like, next I can show how to prove the ***a priori estimate***

$$\|u\|_{H_0^1(U)} \leq C\|f\|_{L^2(U)},$$

which follows immediately from the same argument and is often asked on exams.

5. Sobolev embedding estimate

Show that if $U \subset \mathbb{R}^n$ is bounded and $n > 2$,

$$H_0^1(U) \hookrightarrow L^{2^*}(U), \qquad 2^* = \frac{2n}{n-2}.$$

Skills tested:

- Sobolev inequality,
- critical exponent.

We want to prove the ***Sobolev embedding***

$$H_0^1(U) \hookrightarrow L^{2^*}(U), \qquad 2^* = \frac{2n}{n-2},$$

for a bounded domain $U \subset \mathbb{R}^n$, $n > 2$.

That means there exists a constant $C > 0$ such that

$$\|u\|_{L^{2^*}(U)} \leq C\|u\|_{H_0^1(U)} \qquad \forall u \in H_0^1(U).$$

This is the classical ***Sobolev inequality***, one of the central results in Functional Analysis.

Step 1: Sobolev inequality on $\mathbb{R}^n$

The key result is: If $u \in C_c^\infty(\mathbb{R}^n)$, then

$$\|u\|_{L^{2^*}(\mathbb{R}^n)} \leq C\|\nabla u\|_{L^2(\mathbb{R}^n)},$$

where

$$2^* = \frac{2n}{n-2}.$$

This is the Sobolev inequality on $\mathbb{R}^n$.

Step 2: Extend functions by zero

Take $u \in H_0^1(U)$.

Extend u by zero outside U:

$$\tilde{u}(x) = \begin{cases} u(x), & x \in U, \\ 0, & x \notin U. \end{cases}$$

Then

$$\tilde{u} \in H^1(\mathbb{R}^n),$$

and

$$\|\tilde{u}\|_{L^{2^*}(\mathbb{R}^n)} = \|u\|_{L^{2^*}(U)}.$$

Also,

$$\|\nabla \tilde{u}\|_{L^2(\mathbb{R}^n)} = \|\nabla u\|_{L^2(U)}.$$

Applying the Sobolev inequality in $\mathbb{R}^n$,

$$\|u\|_{L^{2^*}(U)} \leq C\|\nabla u\|_{L^2(U)}.$$

Step 3: Relate to the H_0^1-norm

For $H_0^1(U)$, the norm

$$\|u\|_{H_0^1(U)}$$

is equivalent to

$$\|\nabla u\|_{L^2(U)}$$

by the ***Poincaré inequality***:

$$\|u\|_{L^2(U)} \leq C\|\nabla u\|_{L^2(U)}.$$

Thus,

$$\|u\|_{H_0^1(U)} \sim \|\nabla u\|_{L^2(U)}.$$

Hence,

$$\|u\|_{L^{2^*}(U)} \leq C\|u\|_{H_0^1(U)}.$$

This proves the embedding.

Final result

We have shown that for all $u \in H_0^1(U)$,

$$\boxed{\|u\|_{L^{2^*}(U)} \leq C\|u\|_{H_0^1(U)}, \qquad 2^* = \frac{2n}{n-2}.}$$

Therefore,

$$\boxed{H_0^1(U) \hookrightarrow L^{2^*}(U).}$$

Why $2^* = \frac{2n}{n-2}$?

This exponent is the ***critical Sobolev exponent***. It is the largest exponent for which first derivatives in L^2 force the function itself to lie in L^p.

The relation comes from scaling:

$$\frac{1}{2^*} = \frac{1}{2} - \frac{1}{n}.$$

That gives

$$2^* = \frac{2n}{n-2}.$$

Key exam takeaway

To prove Sobolev embedding on a bounded domain:

1. extend u by zero,
2. apply Sobolev inequality on $\mathbb{R}^n$,
3. use Poincaré inequality.

This is the standard graduate PDE proof structure.

If you'd like, next I can show the ***compact embedding***

$$H_0^1(U) \hookrightarrow\hookrightarrow L^2(U),$$

which is the Rellich–Kondrachov theorem and appears constantly on qualifying exams.

6. Weak compactness + passing to the limit

Suppose u_j solves

$$-\Delta u_j = f_j, \qquad u_j \in H_0^1(U),$$

where $f_j \rightharpoonup f$ in $L^2(U)$, and u_j is bounded in $H_0^1(U)$.

Show a subsequence converges weakly to a weak solution of

$$-\Delta u = f.$$

Skills tested:

- weak convergence,
- compactness,
- limit in weak formulation.

We are given:

- $u_j \in H_0^1(U)$ solves
$$-\Delta u_j = f_j,$$
- $f_j \rightharpoonup f$ weakly in $L^2(U)$,
- (u_j) is bounded in $H_0^1(U)$.

We want to show that some subsequence u_{j_k} converges weakly in $H_0^1(U)$ to a weak solution u of

$$-\Delta u = f.$$

This is a standard ***weak compactness + passage to the limit*** argument in Functional Analysis.

Step 1: Use boundedness in $H_0^1(U)$

Since (u_j) is bounded in $H_0^1(U)$, and $H_0^1(U)$ is a Hilbert space, bounded sets are weakly precompact.

Therefore there exists a subsequence u_{j_k} and some $u \in H_0^1(U)$ such that

$$u_{j_k} \rightharpoonup u \qquad \text{weakly in } H_0^1(U).$$

That means:

$$\int_U \nabla u_{j_k} \cdot \nabla v \, dx \to \int_U \nabla u \cdot \nabla v \, dx$$

for every $v \in H_0^1(U)$.

Step 2: Write the weak formulation for u_j

Since u_j solves

$$-\Delta u_j = f_j,$$

its weak formulation is:

$$\int_U \nabla u_j \cdot \nabla v \, dx = \int_U f_j v \, dx \qquad \forall v \in H_0^1(U).$$

For the subsequence:

$$\int_U \nabla u_{j_k} \cdot \nabla v \, dx = \int_U f_{j_k} v \, dx.$$

Step 3: Pass to the limit on the left-hand side

Since

$$u_{j_k} \rightharpoonup u \quad \text{in } H_0^1(U),$$

we have

$$\int_U \nabla u_{j_k} \cdot \nabla v \, dx \to \int_U \nabla u \cdot \nabla v \, dx.$$

Step 4: Pass to the limit on the right-hand side

Since

$$f_{j_k} \rightharpoonup f \quad \text{in } L^2(U),$$

we know

$$\int_U f_{j_k} \phi \, dx \to \int_U f \phi \, dx \qquad \forall \phi \in L^2(U).$$

Because $v \in H_0^1(U) \subset L^2(U)$, we may take $\phi = v$, giving:

$$\int_U f_{j_k} v \, dx \to \int_U f v \, dx.$$

Step 5: Identify the limit equation

Taking limits in

$$\int_U \nabla u_{j_k} \cdot \nabla v \, dx = \int_U f_{j_k} v \, dx,$$

we obtain

$$\int_U \nabla u \cdot \nabla v \, dx = \int_U f v \, dx \qquad \forall v \in H_0^1(U).$$

This is exactly the weak formulation of

$$-\Delta u = f, \qquad u \in H_0^1(U).$$

Hence u is a weak solution.

Final conclusion

There exists a subsequence u_{j_k} and a function $u \in H_0^1(U)$ such that

$$u_{j_k} \rightharpoonup u \quad \text{weakly in } H_0^1(U),$$

and u satisfies

$$\boxed{\int_U \nabla u \cdot \nabla v \, dx = \int_U f v \, dx \qquad \forall v \in H_0^1(U),}$$

that is,

$$\boxed{-\Delta u = f}$$

in the weak sense.

Key exam structure

This is the standard 3-step compactness argument:

1. Bounded sequence in H_0^1

$$u_j \text{ bounded} \Rightarrow u_{j_k} \rightharpoonup u$$

2. Write weak formulation

$$a(u_j, v) = L_j(v)$$

3. Pass to the limit Use weak convergence on both sides to get

$$a(u, v) = L(v)$$

This exact structure appears constantly in graduate PDE exams.

If you'd like, next I can show how to strengthen this result to ***strong convergence in*** $L^2(U)$ using the ***Rellich compactness theorem***, which is often the next step on comprehensive exams.

7. Maximum principle

Let u solve

$$u_t - \Delta u = 0$$

with $u(x,0) = g(x)$.

Show:

$$\sup_{U\times[0,T]} u \le \sup_U g.$$

Skills tested:

- parabolic maximum principle,
- uniqueness,
- L^∞ bounds.

We want to prove the ***maximum principle*** for the heat equation:

If u solves

$$u_t - \Delta u = 0 \qquad \text{in } U \times (0,T),$$

with initial data

$$u(x,0) = g(x),$$

then

$$\sup_{U\times[0,T]} u \le \sup_U g.$$

This is the standard ***parabolic maximum principle***, one of the central tools in Partial Differential Equations.

Step 1: Define the maximum level

Let

$$M = \sup_U g.$$

We want to show:

$$u(x,t) \le M \qquad \forall (x,t) \in U \times [0,T].$$

Define

$$v(x,t) = u(x,t) - M.$$

Then v satisfies

$$v_t - \Delta v = 0,$$

and

$$v(x,0) = g(x) - M \le 0.$$

So initially, $v \le 0$.

We must show $v \le 0$ for all time.

Step 2: Suppose v becomes positive

Assume for contradiction that $v > 0$ somewhere in $U \times (0, T]$.

Then v attains a positive maximum at some point

$$(x_0, t_0) \in U \times (0, T].$$

At this interior maximum:

$$\nabla v(x_0, t_0) = 0$$
$$\Delta v(x_0, t_0) \leq 0,$$

and since t_0 is the first time of maximum,

$$v_t(x_0, t_0) \geq 0.$$

Thus,

$$v_t - \Delta v \geq 0.$$

But since v satisfies the heat equation,

$$v_t - \Delta v = 0.$$

Hence at the maximum point,

$$0 = v_t - \Delta v \geq 0.$$

This forces equality:

$$v_t = 0, \qquad \Delta v = 0.$$

By the strong maximum principle, if a positive interior maximum occurs, then v must be constant.

But $v(x, 0) \leq 0$, so v cannot be a positive constant.

Contradiction.

Thus $v \leq 0$.

Step 3: Conclude the estimate

Since $v = u - M \leq 0$,

$$u(x, t) \leq M.$$

Therefore,

$$u(x, t) \leq \sup_U g \qquad \forall (x, t) \in U \times [0, T].$$

Hence

$$\boxed{\sup_{U \times [0,T]} u \leq \sup_U g.}$$

Final result

We have shown the ***maximum principle***:

$$\boxed{\sup_{U \times [0,T]} u \leq \sup_U g.}$$

This means the heat equation cannot create temperatures larger than the initial maximum.

Interpretation

For the heat equation,

$$u_t - \Delta u = 0,$$

heat diffuses from hot regions to cold ones.

So:

- peaks flatten out,
- no new maxima are created,
- the maximum temperature decreases or stays constant.

That is exactly what the maximum principle says.

Exam insight

For heat equation maximum principle problems, the standard strategy is:

1. let $M = \sup g$,
2. define $v = u - M$,
3. show $v \leq 0$ by contradiction using an interior maximum.

This is the standard maximum principle template for parabolic PDE.

If you'd like, next I can show how the same argument proves the ***minimum principle***

$$\inf_{U \times [0,T]} u \geq \inf_U g,$$

and combining them gives the useful estimate

$$\|u(\cdot, t)\|_{L^\infty(U)} \leq \|g\|_{L^\infty(U)}.$$

8. Direct method in the calculus of variations

Show that the infimum

$$\inf_{u \in H_0^1(U)} \left(\int_U |\nabla u|^2 \, dx - \int_U f u \, dx \right)$$

is attained.

Skills tested:

- minimizing sequence,
- weak convergence,
- weak lower semicontinuity.

We want to show that the functional

$$J(u) = \int_U |\nabla u|^2 \, dx - \int_U f u \, dx$$

attains its minimum on $H_0^1(U)$, i.e. that there exists $u_0 \in H_0^1(U)$ such that

$$J(u_0) = \inf_{u \in H_0^1(U)} J(u).$$

This is the standard ***direct method in the calculus of variations***, one of the foundational techniques in Calculus of Variations.

Step 1: Define the infimum

Let

$$m = \inf_{u \in H_0^1(U)} J(u).$$

Choose a minimizing sequence $(u_j) \subset H_0^1(U)$ such that

$$J(u_j) \to m.$$

So

$$\int_U |\nabla u_j|^2 \, dx - \int_U f u_j \, dx \to m.$$

We want to show (u_j) has a weakly convergent subsequence.

Step 2: Show the minimizing sequence is bounded

We estimate the linear term using Cauchy–Schwarz:

$$\left| \int_U f u_j \, dx \right| \leq \|f\|_{L^2(U)} \|u_j\|_{L^2(U)}.$$

Using Poincaré:

$$\|u_j\|_{L^2(U)} \leq C \|\nabla u_j\|_{L^2(U)}.$$

Hence

$$\left| \int_U f u_j \, dx \right| \leq C \|f\|_{L^2(U)} \, \|\nabla u_j\|_{L^2(U)}.$$

Thus

$$J(u_j) \geq \|\nabla u_j\|_{L^2}^2 - C\|f\|_{L^2} \, \|\nabla u_j\|_{L^2}.$$

Let

$$t_j = \|\nabla u_j\|_{L^2}.$$

Then

$$J(u_j) \geq t_j^2 - C t_j.$$

Since $J(u_j) \to m$, the values $J(u_j)$ are bounded above, so $t_j^2 - C t_j$ is bounded above, which implies t_j is bounded.
Therefore,

$$(u_j) \text{ is bounded in } H_0^1(U).$$

Step 3: Extract a weakly convergent subsequence

Since $H_0^1(U)$ is reflexive, there exists a subsequence (still denoted u_j) and some $u_0 \in H_0^1(U)$ such that

$$u_j \rightharpoonup u_0 \qquad \text{weakly in } H_0^1(U).$$

Step 4: Use weak lower semicontinuity

The map

$$u \mapsto \int_U |\nabla u|^2 \, dx$$

is weakly lower semicontinuous, so

$$\int_U |\nabla u_0|^2 \, dx \leq \liminf_{j \to \infty} \int_U |\nabla u_j|^2 \, dx.$$

Also, the linear functional

$$u \mapsto \int_U f u \, dx$$

is weakly continuous, so

$$\int_U f u_j \, dx \to \int_U f u_0 \, dx.$$

Therefore,

$$J(u_0) = \int_U |\nabla u_0|^2 \, dx - \int_U f u_0 \, dx$$

satisfies

$$J(u_0) \leq \liminf_{j \to \infty} J(u_j).$$

Since $J(u_j) \to m$,

$$J(u_0) \leq m.$$

But m is the infimum, so

$$J(u_0) = m.$$

Thus the infimum is attained.

Final conclusion

There exists $u_0 \in H_0^1(U)$ such that

$$\boxed{J(u_0) = \inf_{u \in H_0^1(U)} \left(\int_U |\nabla u|^2 \, dx - \int_U f u \, dx \right).}$$

So the minimization problem has a minimizer.

Why this works

The direct method always uses:

- 1. Minimize:
  Take a minimizing sequence.
- 2. Compactness:
  Show boundedness, then extract weak convergence.
- 3. Lower semicontinuity:
  Pass to the limit.

This gives existence of a minimizer.

PDE interpretation

The minimizer u_0 is the weak solution of

$$-\Delta u = \frac{f}{2}$$

(up to constants in the functional).

Thus the variational problem corresponds to an elliptic PDE.

This variational viewpoint is central in Partial Differential Equations.

—

If you'd like, next I can show how to derive the ***Euler–Lagrange equation*** of this minimization problem and recover the PDE satisfied by the minimizer.

9. Fundamental solution of the heat equation

Verify that

$$\Phi(x,t) = \frac{1}{(4\pi t)^{n/2}} e^{-|x|^2/4t}$$

solves the heat equation.

Skills tested:

- direct differentiation,
- fundamental solutions.

10. Fourier transform solution of the heat equation

Solve

$$u_t - \Delta u = 0, \qquad u(x,0) = g(x)$$

on $\mathbb{R}^n$ using the Fourier transform.

Show:

$$u(x,t) = \Phi(\cdot,t) * g(x).$$

Skills tested:

- Fourier transform,
- ODE in frequency space,
- convolution.

Why these 10 matter

These ten problems cover the most common exam strategies:

Energy methods
Problems 1,2,7
These test uniqueness and estimates.

Weak solutions
Problems 3,4,6
These test Sobolev space fluency.

Sobolev/compactness
Problems 5,6,8
These test existence methods.

Explicit formulas
Problems 9,10
These test computational PDE techniques.

If I were prioritizing for exams:

Study these in order:

1. ***Energy estimates***
2. ***Weak formulations***
3. ***Lax–Milgram***
4. ***Sobolev embeddings***
5. ***Compactness arguments***
6. ***Maximum principle***
7. ***Fourier methods***

That sequence covers a huge percentage of PDE comprehensive exams.

If you'd like, next I can create a ***mock PDE comprehensive exam (3 problems, 3 hours)*** at the same level as a graduate qualifying exam.